

Does Gravity See Slow Mantle Regions as Heavy or Light?

A Global-Scale Comparison of Mantle Models with Observed Gravity

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Abstract

A central issue in the current LLSVP debate is whether slow seismic provinces should be treated as intrinsically dense structures, a premise often used to argue that they should remain near the equator rather than occupy a non-equatorial degree-2 configuration. This study tests a narrower observational question: at Earth’s global degree-2, or quadrupole, scale, does the observed static-gravity field better match the commonly adopted mapping in which slow seismic anomalies act as denser mass anomalies, or the globally reversed mapping in which they act as lighter anomalies? Using a reference-compatible central gravity residual, seven of eight full-mantle model fields favored the globally reversed sign. Six of the seven velocity-based models therefore favored the slow-as-light mapping, while the remaining reversed result came from a native density-field model. All eight model fields retained their deterministic sign classification in the 10,000 realizations under the documented formal gravity-coefficient covariance: seven remained reversed and one remained adopted. These results do not support treating the assumption that LLSVPs are intrinsically denser than the surrounding mantle as settled evidence against a non-equatorial geometry. They also do not establish that LLSVP material is intrinsically buoyant or that a non-equatorial degree-2 configuration would be dynamically stable. The finding is limited to a large-scale static-gravity comparison: at degree 2, the observed gravity field more often favored the sign in which slow seismic anomalies behave as relatively lighter mass anomalies. It does not directly determine LLSVP density, composition, rheology, or behavior during mantle displacement.

Keywords: seismic tomography; gravity field; degree 2; sign convention; mantle density; covariance; principal axes

1 Background

Seismic tomography primarily images variations in seismic wave speed. Density enters only through additional physical assumptions, joint inversions, mineral-physics relations, or independently supplied density fields [1–3]. A slow anomaly can therefore acquire different density interpretations depending on whether temperature, composition, phase, attenuation, anisotropy, and reference-state effects are treated explicitly. The velocity map does not arrive with a density sign attached.

That distinction matters whenever a tomographic field is converted into a gravity or inertia tensor. A sign choice determines whether a slow seismic anomaly contributes as a positive or negative mass anomaly. It also determines whether a fixed eigendirection is interpreted as a minimum- or maximum-moment direction. The geometry and the physical role are related, but they are not interchangeable.

Rather than settle the sign by convention, I asked the observed gravity field which mapping it preferred. Gravity gets a vote.

The test is deliberately narrow. It compares full-mantle degree-2 model vectors with a declared reference-compatible static-gravity residual. It does not infer a spatially uniform density for the large low-shear-velocity provinces (LLSVPs), and it does not replace thermodynamic or mineral-physics analysis. Its purpose is to identify the sign that better matches the observed large-scale coefficient geometry under the stated residual and covariance assumptions.

2 Model setup

2.1 Observed degree-2 gravity residual

The coefficient vector is ordered as

$$\mathbf{d} = [C_{20}, C_{21}, S_{21}, C_{22}, S_{22}]^T. \quad (1)$$

The primary branch, `R_compatible_central`, was derived from GOCO2025s by subtracting the declared IERS reference figure and the validated compatible mean-ocean, grounded-Antarctic-ice, and grounded-Greenland-ice coefficient vectors. The coefficients are fully normalized, zero tide, and referenced to a radius of 6.378×10^6 m. The observed gravity coefficients were taken from GOCO2025s [5]. The axisymmetric reference figure follows IERS Conventions (2010) [4].

This branch is explicitly partial. It does not include a reference-compatible LITHO1.0 subtraction, topographic or crustal compensation closure, atmosphere, hydrology, mountain glaciers, glacial isostatic adjustment, or a complete covariance for the correction vectors. Table 1 lists the central residual and the formal coefficient uncertainties used in the fit.

2.2 Full-mantle model fields

Eight full-mantle fields were evaluated: S40RTS, TX2011, SEISGLOB2, SPani, SEMUCB-WM1, CSEM2-Globe-RHO, CSEM2-Globe-VSV, and UNICA25-absolute-Vp. These span global S-wave tomography, anisotropic S-wave

Tomography-to-gravity sign-convention test

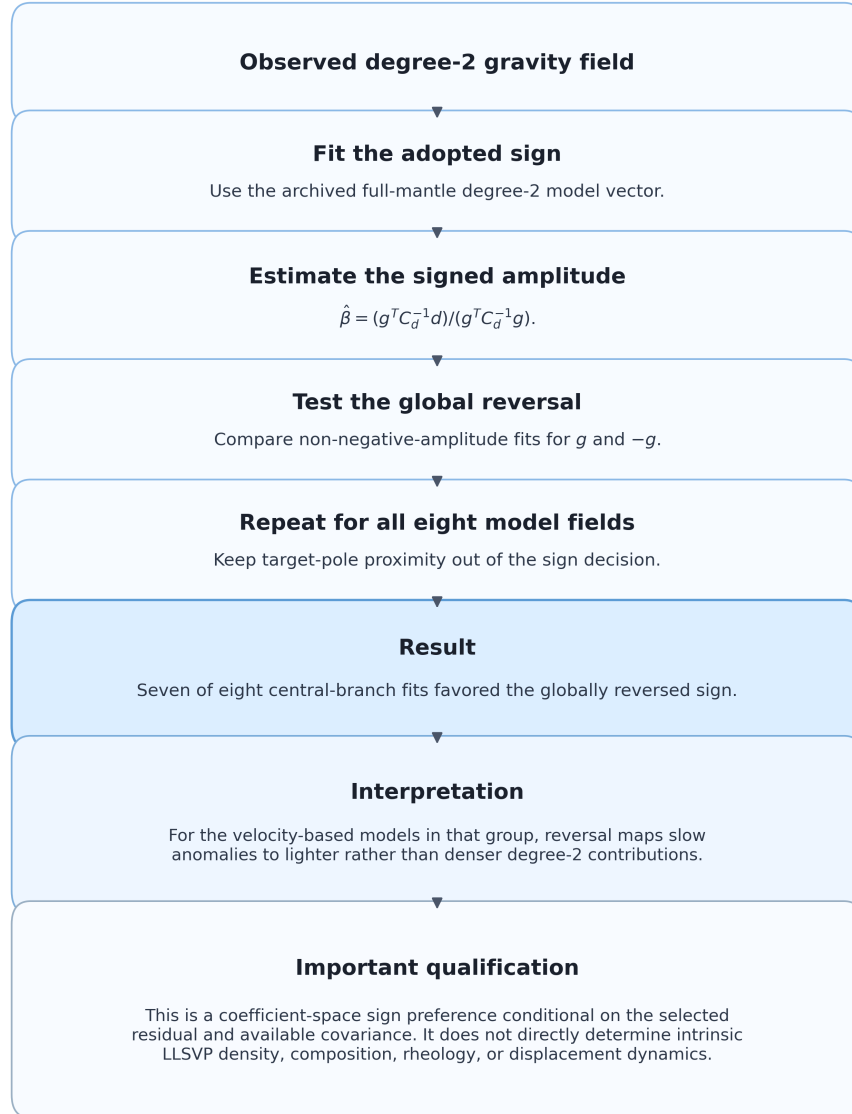


Figure 1: Workflow for the sign-convention test. The sign decision is made entirely in degree-2 coefficient space. Target-pole proximity is excluded from the fit, so the test does not select a sign because it moves a preferred axis toward a desired location. The final box states the central limitation: this is a conditional sign preference rather than a direct density or dynamics measurement.

Table 1: Primary residual coefficients and formal standard deviations. The coefficient uncertainties come from the archived reference-figure gravity product. Uncertainty in the compatible correction vectors is not included.

Coefficient	Central residual	Formal σ
C_{20}	6.055629264e-06	2.000019572e-11
C_{21}	5.795538126e-06	8.775592797e-14
S_{21}	4.910665758e-06	9.787141450e-14
C_{22}	-3.803033039e-06	1.359212511e-13
S_{22}	-3.376437631e-06	1.298328942e-13

tomography, a native density-related field, a vertical-shear-velocity field, and a depth-referenced P-wave anomaly product [6–12].

Seven entries are velocity-based. Their adopted conversion used a multiplier of -1 , corresponding to the project’s original slow-as-dense proxy. CSEM2-Globe-RHO is different: it is a native density-related field whose adopted sign was retained. This distinction becomes important when translating the seven-of-eight model-field result into plain language.

For each model m , the archived degree-2 prediction is

$$\mathbf{g}_m = [g_{20}, g_{21}, g_{S21}, g_{22}, g_{S22}]^T. \quad (2)$$

The model vectors are shape vectors. Their normalization changes the fitted amplitude but cannot change the sign preference, because positive rescaling of $\mathbf{g}_m$ rescales $\hat{\beta}$ inversely while preserving $\hat{\beta}\mathbf{g}_m$.

2.3 Signed coefficient-space fit

The observed residual is fit as

$$\mathbf{d} \approx \beta \mathbf{g}_m. \quad (3)$$

With $\mathbf{W} = \mathbf{C}_d^{-1}$, the unregularized weighted least-squares estimator is

$$\hat{\beta}_m = \frac{\mathbf{g}_m^T \mathbf{W} \mathbf{d}}{\mathbf{g}_m^T \mathbf{W} \mathbf{g}_m}, \quad (4)$$

and its formal standard error is

$$\sigma_{\beta,m} = (\mathbf{g}_m^T \mathbf{W} \mathbf{g}_m)^{-1/2}. \quad (5)$$

The sign rule is direct:

$$\hat{\beta}_m > 0 \Rightarrow \text{adopted sign}, \quad (6)$$

$$\hat{\beta}_m < 0 \Rightarrow \text{globally reversed sign}. \quad (7)$$

A fit is treated as indeterminate when $|\hat{\beta}| \leq 2\sigma_\beta$ or the formal 95% interval crosses zero. No regularization term was used in this branch.

The two constrained hypotheses use a non-negative amplitude under each sign:

$$a_{\text{adopted}} = \max(0, \hat{\beta}), \quad (8)$$

$$a_{\text{reversed}} = \max(0, -\hat{\beta}). \quad (9)$$

Their weighted residuals are

$$\chi_{\text{adopted}}^2 = (\mathbf{d} - a_{\text{adopted}}\mathbf{g})^T \mathbf{W} (\mathbf{d} - a_{\text{adopted}}\mathbf{g}), \quad (10)$$

$$\chi_{\text{reversed}}^2 = (\mathbf{d} + a_{\text{reversed}}\mathbf{g})^T \mathbf{W} (\mathbf{d} + a_{\text{reversed}}\mathbf{g}). \quad (11)$$

Thus, $\Delta\chi^2 = \chi_{\text{adopted}}^2 - \chi_{\text{reversed}}^2 > 0$ favors reversal. This constrained comparison avoids the trivial identity between an unrestricted fit to $\mathbf{g}$ and an unrestricted fit to $-\mathbf{g}$.

2.4 Covariance and Monte Carlo propagation

The available covariance is diagonal:

$$\mathbf{C}_d = \text{diag}(\sigma_{C_{20}}^2, \sigma_{C_{21}}^2, \sigma_{S_{21}}^2, \sigma_{C_{22}}^2, \sigma_{S_{22}}^2). \quad (12)$$

Ten thousand realizations were drawn:

$$\mathbf{d}^{(r)} \sim \mathcal{N}(\mathbf{d}, \mathbf{C}_d), \quad (13)$$

with the compatible correction vectors held fixed. The resulting intervals therefore quantify only the documented formal coefficient uncertainty. They are not a complete uncertainty budget for the mantle residual.

2.5 Role interpretation under sign reversal

Let the trace-free degree-2 tensor be $\mathbf{Q}$ with

$$\mathbf{Q}\mathbf{v}_i = \lambda_i \mathbf{v}_i. \quad (14)$$

A global sign reversal gives

$$(-\mathbf{Q})\mathbf{v}_i = (-\lambda_i)\mathbf{v}_i. \quad (15)$$

The eigenvectors are unchanged. The eigenvalue ordering reverses, so the minimum and maximum roles exchange while the intermediate role remains intermediate. Figure 2 illustrates this algebraic result.

2.6 Validation workflow

The standalone public reproduction script executed successfully using the ten frozen files under data/ and validation/. It reproduced the archived deterministic estimates, formal errors, constrained comparisons, and Monte Carlo summaries and recorded SHA-256 hashes for those ten public inputs.

This package supports reproduction of the sign test from its frozen degree-2 inputs. It provides the observed gravity-residual coefficients, the eight model-prediction vectors, the documented formal coefficient covariance, the fitting and decision rules, the Monte Carlo settings, and the standalone code required to reproduce the reported beta estimates, residual comparisons, and sign classifications. It does not reproduce the earlier processing steps used to derive those degree-2 vectors from the original three-dimensional tomography models. A complete end-to-end reconstruction would additionally require the raw tomography grids, original gravity products, shallow-correction inputs, model parsers, preprocessing notebooks, software environment, and full upstream provenance.

Role swap under global tomography sign reversal



Global sign reversal leaves the three axes in place, swaps min/max roles, and leaves mid as mid.

Figure 2: Global sign reversal changes eigenvalue ordering without rotating the eigendirections. A location identified by the minimum eigenvalue under the adopted sign becomes the maximum-eigenvalue direction under reversal; the intermediate direction remains intermediate. This role swap is exact apart from floating-point eigensolver precision.

3 Results

3.1 Deterministic sign fits

Figure 3 shows the fitted signed amplitudes. Seven model fields lie on the negative side of zero. CSEM2-Globe-VSV is the single positive central-branch result.

The central result is:

For seven of the eight full-mantle model fields, the global-scale static-gravity comparison favored the globally reversed sign. Among the seven velocity-based fields, six favored the mapping in which slow seismic anomalies contribute as relatively lighter rather than denser mass anomalies.

Strictly, one of the seven reversed entries is CSEM2-Globe-RHO, a native density-related field rather than a velocity model. Among the seven velocity-based entries, six favor the slow-as-light mapping and CSEM2-Globe-VSV favors the adopted slow-as-dense mapping. The quoted statement is therefore a compact description of the model-field result, not a claim that all eight inputs measure seismic velocity.

The observed degree-2 gravity field more often preferred the opposite tomography sign.

Table 2 gives the model-level results. All seven negative central-branch values are many formal standard errors below zero under the available covariance, while the CSEM2-Globe-VSV value is many formal standard errors above zero.

3.2 Constrained residual comparison

The constrained residual comparison yields the same sign classification. Figure 4 shows that the reversed sign has the smaller weighted residual for seven model fields. CSEM2-Globe-VSV is the exception.

3.3 Monte Carlo sign stability

Figure 5 displays the 10,000-realization distributions. Seven model fields produced $\beta < 0$ in every saved realization, while CSEM2-Globe-VSV produced $\beta > 0$ in every realization. The apparent decisiveness reflects the narrow formal gravity-coefficient uncertainties relative to the fitted amplitudes.

The correct reading is therefore specific: the sign is stable to the formal gravity-coefficient perturbations that were actually propagated. “100% of realizations” does not mean 100% physical confidence.

3.4 Residual and coefficient sensitivity

The count remains seven reversed and one adopted under all three declared residual branches: the reference-figure residual, the reference-compatible central partial residual, and the LITHO1 anomaly-only sensitivity branch. The identity of the exception is not invariant. UNICA25-absolute-Vp is the adopted result under the reference residual, whereas CSEM2-Globe-VSV is the adopted result under the central and anomaly-sensitivity residuals.

The leave-one-coefficient-out test is more revealing. Omitting C_{21} or S_{21} yields eight reversed results. Omitting C_{22} yields six reversed and two adopted. Omitting S_{22} yields an even split: four reversed and four adopted. Figure 6 and Table 4 make this dependence explicit.

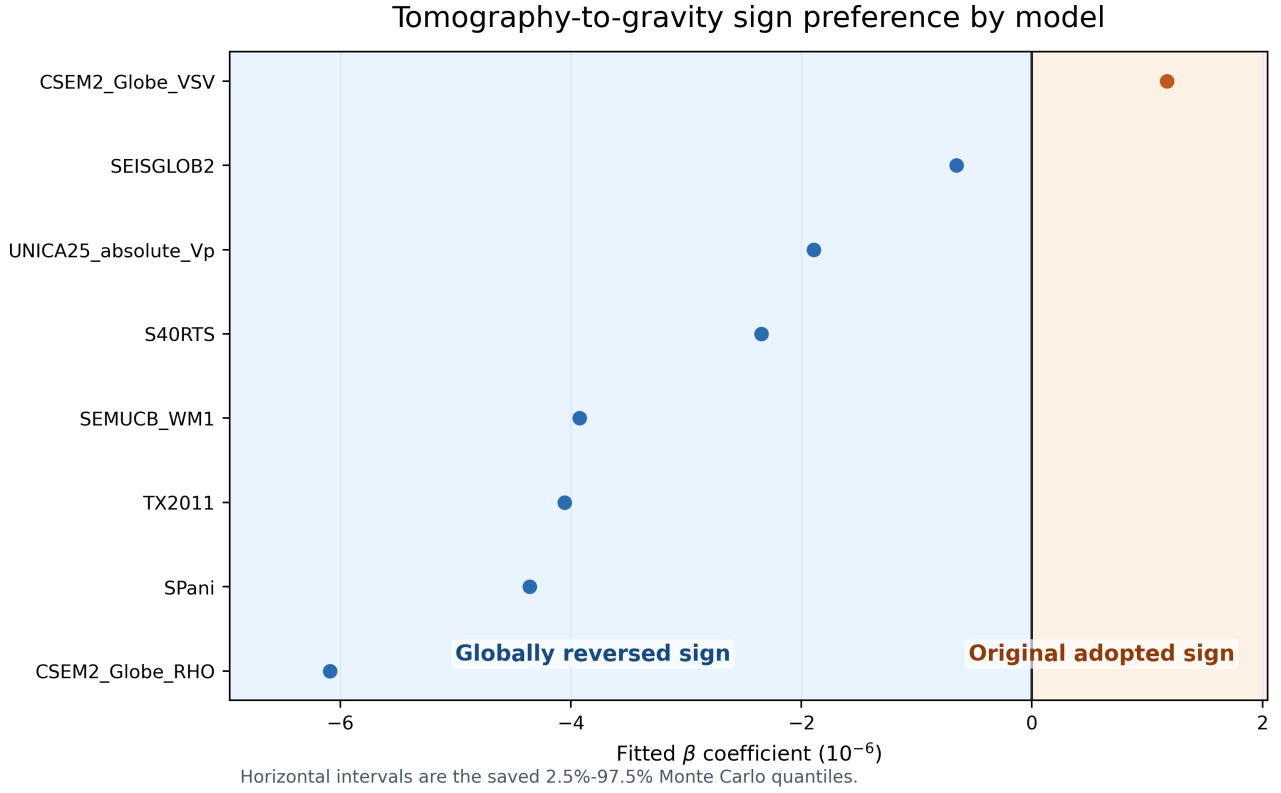


Figure 3: Signed β estimates for the primary reference-compatible residual. Negative values support a global reversal of the adopted model sign; positive values support the adopted sign. The horizontal intervals are the saved 2.5–97.5% Monte Carlo quantiles and are visually small because they propagate only the documented formal gravity-coefficient covariance.

Table 2: Model-by-model sign results for the primary residual. β and the Monte Carlo interval are reported in units of 10^{-6} . “Fraction $\beta < 0$ ” refers only to the documented formal covariance with correction vectors held fixed.

Model	Input quantity	$\hat{\beta}$	MC 95% interval	Fraction $\beta < 0$	Preferred	Model-level interpretation
S40RTS	V_S perturbation	-2.347340	[-2.347340, -2.347339]	100.0%	Reversed	slow-as-light
TX2011	V_S perturbation	-4.054710	[-4.054711, -4.054710]	100.0%	Reversed	slow-as-light
SEISGLOB2	V_S perturbation	-0.656103	[-0.656103, -0.656102]	100.0%	Reversed	slow-as-light
SPani	anisotropic V_S perturbation	-4.357798	[-4.357798, -4.357798]	100.0%	Reversed	slow-as-light
SEMUCB_WM1	V_S perturbation	-3.923406	[-3.923406, -3.923406]	100.0%	Reversed	slow-as-light
CSEM2_Globe_RHO	native density-related field	-6.089010	[-6.089010, -6.089009]	100.0%	Reversed	reversed native density-field sign
CSEM2_Globe_VSV	V_{SV} -related field	1.172924	[1.172924, 1.172925]	0.0%	Adopted	slow-as-dense
UNICA25_absolute_Vp	depth-mean V_P anomaly	-1.892465	[-1.892466, -1.892465]	100.0%	Reversed	slow-as-light

Table 3: Constrained weighted residual norms. Values are shown in units of 10^7 . The percent change is $(r_{\text{adopted}} - r_{\text{reversed}})/r_{\text{adopted}}$; positive values favor reversal.

Model	Adopted residual	Reversed residual	Change	Favored sign
S40RTS	9.1314	8.9471	+2.019%	Reversed
TX2011	9.1314	8.6835	+4.905%	Reversed
SEISGLOB2	9.1314	9.1167	+0.161%	Reversed
SPani	9.1314	8.5870	+5.962%	Reversed
SEMUCB_WM1	9.1314	8.6643	+5.115%	Reversed
CSEM2_Globe_RHO	9.1314	8.1675	+10.556%	Reversed
CSEM2_Globe_VSV	9.0945	9.1314	-0.405%	Adopted
UNICA25_absolute_Vp	9.1314	9.0400	+1.001%	Reversed

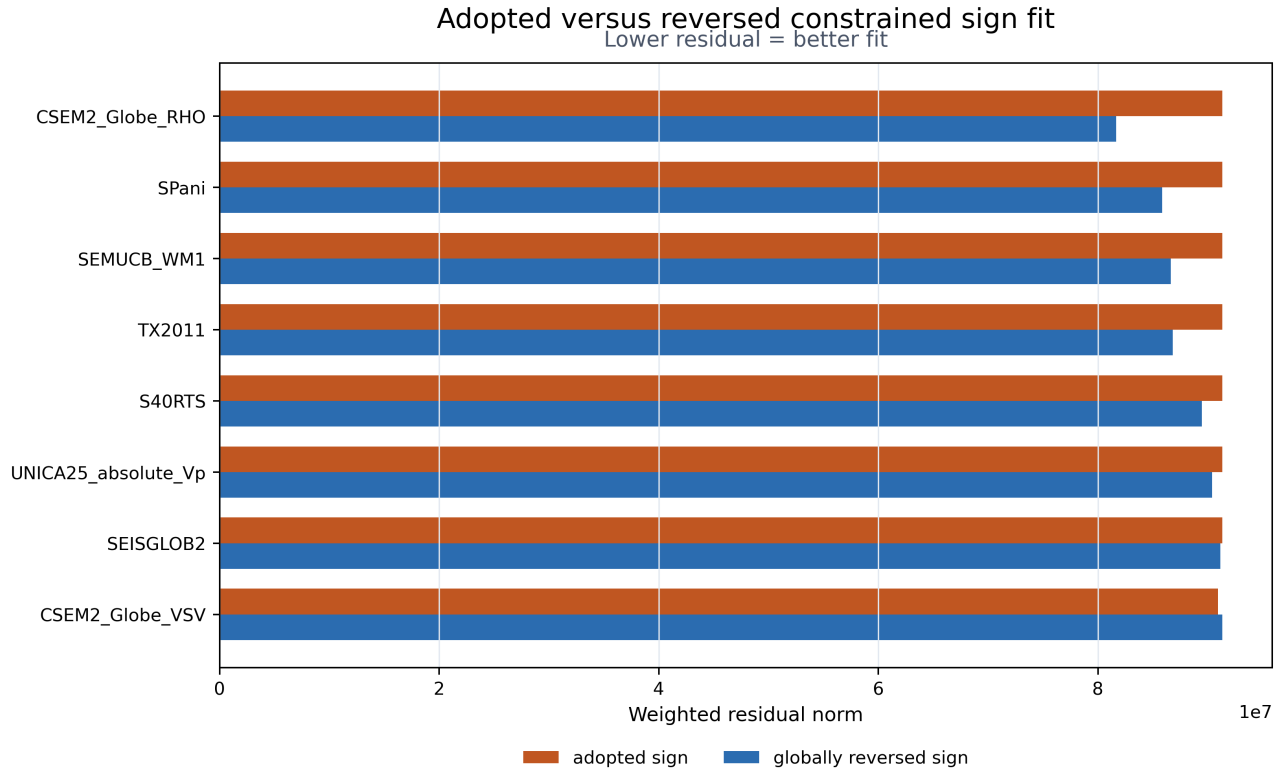


Figure 4: Weighted residual norms for the two non-negative-amplitude sign hypotheses. The reversed sign improves the fit for seven model fields. The largest relative reduction occurs for CSEM2-Globe-RHO; SEISGLOB2 shows the smallest improvement among the reversed cases. The plot compares fit quality under the chosen covariance and should not be read as a complete physical likelihood ratio.

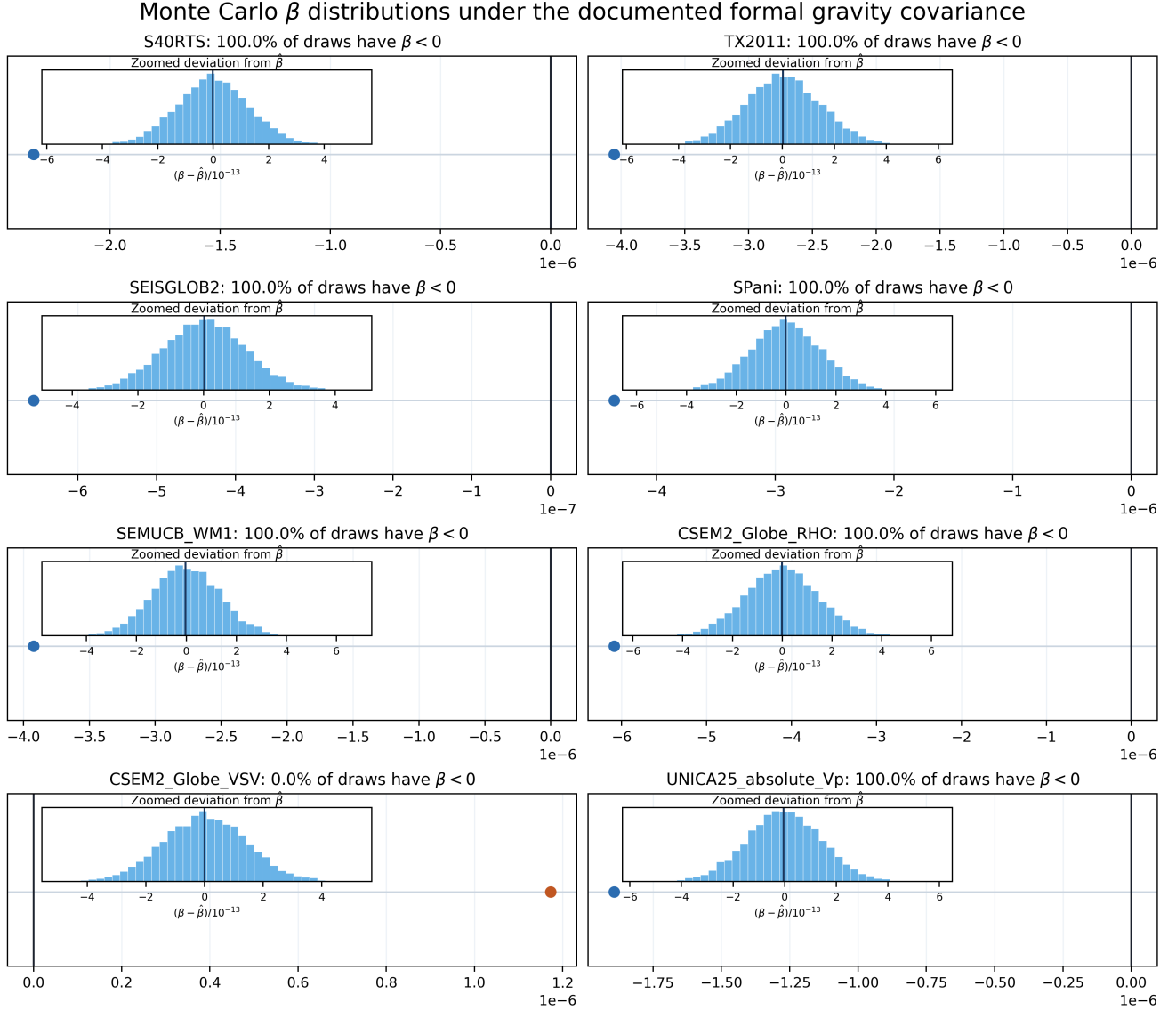


Figure 5: Monte Carlo distributions under the documented diagonal formal covariance. Each main panel places the fitted value and zero on the same horizontal ruler; the inset magnifies deviations around $\hat{\beta}$. The distributions do not include uncertainty in shallow corrections, gravity-reference closure, tomography model structure, or coefficient correlations that were unavailable in the archived inputs.



Figure 6: Sensitivity of the model count to residual definition and coefficient omission. The seven-of-eight count is unchanged across the three declared residual branches, but the result has substantial S_{22} leverage: removing S_{22} reduces the count to four reversed and four adopted.

Table 4: Sign-count sensitivity.

Case	Rev.	Adopt.	Indet.
Reference residual	7	1	0
Central residual	7	1	0
LITHO1 anomaly branch	7	1	0
Omit C_{20}	7	1	0
Omit C_{21}	8	0	0
Omit S_{21}	8	0	0
Omit C_{22}	6	2	0
Omit S_{22}	4	4	0

4 Interpretation

Observation. Seven of eight central-branch model fields have $\hat{\beta} < 0$, and the constrained reversed-sign residual is smaller for the same seven. The model count remains seven of eight across the three declared residual branches.

Interpretation. At degree 2, the observed residual aligns more closely with the negative of the adopted model vector for most of the tested fields. For velocity-based models, that reversal corresponds to slow anomalies contributing with the lighter sign rather than the denser sign in the large-scale gravity comparison.

Physical inference. The result supports a slow-as-light large-scale mapping more often than the adopted slow-as-dense mapping within this coefficient-space experiment. It does not establish a uniform thermodynamic state for the LLSVPs. Local density contrasts can differ from the global degree-2 projection, and a native density field is represented among the eight inputs.

This was a degree-2, coefficient-space sign preference, conditional on the chosen reference-compatible gravity residual and available covariance model.

It was not a direct measurement of the intrinsic density of the LLSVP material, and it was based on the full-mantle model comparison rather than an isolated thermodynamic measurement of the lowermost piles.

This conclusion remains conditional on the available gravity-coefficient uncertainty model. Uncertainty associated with the shallow-mantle corrections was not fully propagated, so the reported confidence intervals do not represent the complete uncertainty budget.

This analysis does not determine:

- intrinsic density,
- composition,
- mantle rheology, or
- displacement dynamics.

The result also does not prove that every slow anomaly is buoyant. It reports the sign of the best-fitting degree-2 projection. Higher-degree structure, radial variation, compensation, chemical heterogeneity, and local departures from the fitted global relation remain outside the test.

5 Discussion

This study addresses the assumed density sign underlying the equatorial-placement objection; it does not independently determine the mechanically stable orientation of the structures.

5.1 What changes

The practical consequence is straightforward: the original adopted tomography-to-density sign should not be treated as automatically preferred at degree 2. The static-gravity residual selected the globally reversed mapping for seven of eight central-branch model fields. The same model count appears in two alternative residual branches, which makes the ensemble-level result harder to dismiss as a single-branch accident.

Global sign reversal leaves the eigendirections fixed while reversing the eigenvalue ordering. A direction identified as the maximum-moment axis under the adopted sign becomes the minimum-moment axis under the reversed sign, and vice versa; the intermediate axis remains intermediate. The geometry stays in place; the labels swap.

5.2 What remains conditional

The formal Monte Carlo result is narrower than the physical uncertainty. The correction vectors were fixed, and their covariance was unavailable. The central residual is partial. The S_{22} omission test changes the count from seven-to-one to four-to-four. These facts require the sign result to remain conditional even though the archived formal errors place each individual β far from zero.

The residual-branch sensitivity also shows that ensemble count and model identity are different robustness questions. Seven of eight is stable across the three branches; the identity of the sole adopted model changes between the reference residual and the two corrected branches.

5.3 The remaining dynamical question

The remaining question is dynamical: whether buoyant lowermost structures would move coherently with the mantle shell, deform relative to the core-mantle boundary, or re-equilibrate after displacement.

This paper does not answer that question. A coefficient-space sign preference does not prescribe mechanical coupling, relaxation time, viscosity, phase behavior, or survival of lowermost-mantle structures during large reorientation. Those questions require a separate dynamical model with explicit forces, constitutive behavior, boundary conditions, and timescales.

The contribution of the present study is narrower and cleaner. It establishes which sign the observed degree-2 gravity field more often supports.

6 Conclusion

This study tested the adopted and globally reversed tomography-to-gravity mappings against a declared degree-2 static-gravity residual. Seven of eight full-mantle model fields favored the globally reversed sign in the primary branch, and the constrained residual comparison gave the same classification. Under the documented formal coefficient covariance, those seven retained $\beta < 0$ in all 10,000

saved realizations.

For the velocity-based models in the reversed group, the preferred mapping treats slow seismic anomalies as lighter rather than denser at degree 2. One reversed entry is a native density-related field, so the result should be described as a seven-of-eight model-field sign preference and a six-of-seven slow-as-light result among the velocity-based models.

The conclusion remains conditional on the reference-compatible residual, the formal diagonal covariance, fixed correction vectors, and an incomplete shallow-correction uncertainty budget. The leave-one-coefficient-out test identifies substantial S_{22} leverage, and the identity of the single adopted model changes under the uncorrected reference residual.

Future work should close the shallow-load covariance budget, test alternative gravity products and reference states, propagate tomography-model uncertainty, examine depth-resolved sign behavior, and then address the separate dynamical question of how lowermost-mantle structures would move, deform, or re-equilibrate.

Disclosure and availability

This is an independent geophysical modeling case study. The public repository contains the manuscript, frozen degree-2 publication inputs, archived validation outputs, and standalone code required to reproduce the reported sign test from those frozen inputs. Private working notebooks, raw three-dimensional tomography volumes, broader cleaned intermediate datasets, and upstream pre-processing pipelines are intentionally excluded. Reproducibility therefore begins with the frozen degree-2 coefficient vectors rather than with the original tomography grids. The public script validates the reconstructed results against the archived outputs and records hashes for the included public inputs.

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